Solar for African SMEs

Does solar pay for itself for small businesses? A Power BI report on 196 cases across 49 African countries.

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THE QUESTION

How long does solar take to pay for itself for a small African business?

196
cases in the data

49
African countries

4
regions: West, East, Central, Southern

4
solar systems, one per business type

Each case is one country and one system, with its price, yearly savings, return and payback. North Africa is not in the data.

The short answer

3.0 years

On average, an SME earns back the full cost of its solar system in about three years, a 34.0% yearly return.

103 of 196 cases pay back in under 3 years

2.1 to 4.4 years from fastest to slowest

BY SYSTEM

The most expensive system pays back fastest

System and business	Price	Saved a year	Return	Payback
Water Pump, agriculture	$1,261	$472	37.4%	2.7 years
Small SHS, food retail	$194	$68	35.1%	2.9 years
Larger SHS, small manufacturing	$572	$187	32.6%	3.1 years
Refrigeration, cold storage	$941	$289	30.7%	3.3 years

Price does not decide payback. The water pump costs the most, but it saves the most each year, so it earns its cost back first.

BY REGION

West Africa pays back five months sooner than Southern Africa



Average return and payback. The water pump leads and refrigeration trails in every region, so the system choice matters more than the place.

BY COUNTRY

Even the lowest country pays back in about four years

Highest return

Country	Return
South Sudan	42.9%
Somalia	41.6%
Central African Rep.	40.6%

Lowest return

Country	Return
Seychelles	24.2%
Mauritius	26.2%
Angola	26.5%

Prices are the same in every country, so these gaps come only from how much a business saves each year.

FINANCIAL BENEFIT

Only 9 of 196 cases are rated moderate

81

Very high

2.6 years to pay back

106

High

3.2 years to pay back

9

Moderate

4.2 years to pay back

The rating follows payback closely: the faster a system pays back, the higher its rating.

WHAT IF

A 20% subsidy cuts payback from 3.0 to 2.4 years

2.4 yrs
payback, from 3.0

42.5%
return, from 34.0%

$148
average subsidy per system

Subsidy needed for a two-year payback

System	Subsidy
Water Pump	25%
Small SHS	30%
Larger SHS	35%
Refrigeration	39%

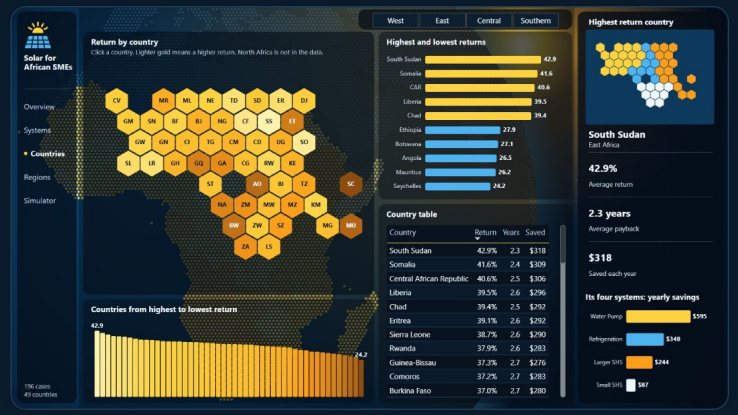
Scenario from the report's simulator, with savings unchanged. The subsidy is a what-if, not part of the data.

THE REPORT

Five pages, each built on a globe of Africa



Systems: click a card to see that system



Countries: a clickable map of 49 countries



Simulator: test a subsidy and see payback change

The Overview and Regions pages complete the set. A region filter carries across pages, and every chart answers to clicks.

WHAT TO DO

Where solar money goes furthest

Lead with water pumps

They return 37.4% and pay back in 2.7 years, the best of the four, for farms.

Use partial subsidies

About 25% brings a water pump to a two-year payback. Full grants are not needed.

Support cold storage

Refrigeration is slowest at 3.3 years and needs the largest subsidy, about 39%.

Fill the data gaps

Add North Africa, local prices and dates, so the report can track change over time.

HOW IT WAS BUILT

Power BI, DAX and Deneb

Part	What it does
Data model	Cases, countries, systems and benefit levels
DAX	58 measures, including the simulator
Deneb	Custom charts and the clickable map
Background	A dotted globe drawn from the data

LIMITS OF THE DATA

No dates. The report shows a snapshot, not a trend.

One price per system. Country differences come only from savings.

No North Africa. Five countries are missing.

Return is yearly savings over price, before maintenance, finance or battery costs.



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github.com/bihbrandysango-cloud/solar-sme-financial-impact

bihbrandysango-cloud.github.io